

# SHA-1 RTL optimization on VTR

Technical report · reproducible open 45 nm FPGA proxy

**Four cycle-equivalent RTL rewrites reduce  
estimated composite PPA by 5.98%**

Prepared by Göther Labs

Evidence snapshot: 5 August 2026

# 1. Executive result

## THE ACCEPTED SOLUTION ONLY

The accepted RTL reduces the equal-weight area-delay-energy score by 5.98% (95% CI: 5.69% to 6.27%) without changing the module interface, state-visible behavior, latency or register count.

| Metric         | Baseline median | Accepted median | Paired improvement | 95% CI             | State   |
|----------------|-----------------|-----------------|--------------------|--------------------|---------|
| Total area     | 16,614,693 MWTa | 16,614,693 MWTa | 0.03%              | +0.09%<br>−0.04%   | neutral |
| Critical path  | 15.0054 ns      | 13.28085 ns     | 11.43%             | +11.98%<br>+10.87% | better  |
| Energy / block | 12.3896 nJ      | 11.6399 nJ      | 6.14%              | +6.52%<br>+5.77%   | better  |
| Composite PPA  | 1.000000        | 0.940188        | 5.98%              | +6.27%<br>+5.69%   | better  |

## Certified PPA improvement vs. baseline

64 fixed, paired VPR seeds · lower metric values are better

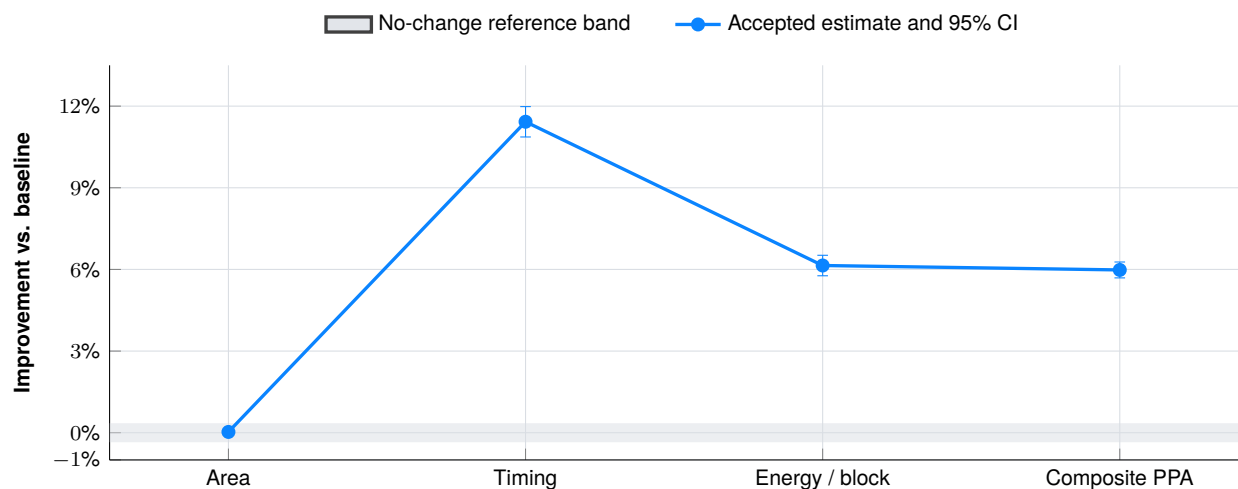


Figure 1. Certified paired improvements. Area is neutral because its interval crosses no change.

All 64 paired seeds favor the accepted RTL for timing, energy and composite score. Area produces 14 wins, 41 ties and 9 losses and is therefore reported as neutral. The result is cumulative for the four-line revision; no exact per-line PPA attribution is claimed.

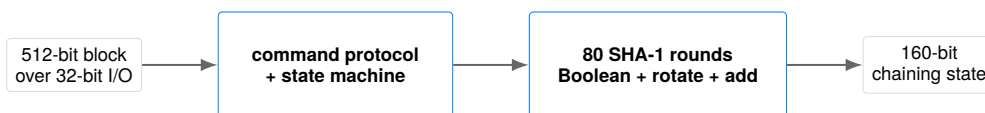
## 2. Module and benchmark

### WHAT IS BEING OPTIMIZED

The evaluated module is a sequential implementation of SHA-1 compression from the public [Verilog-to-Routing repository](#). The exact upstream source is the [pinned sha.v at commit 95f5c6de...](#)

It consumes a 512-bit message block through a 32-bit command/data interface and updates a 160-bit chaining state through 80 rounds of Boolean mixing, rotations, a message schedule and modulo- $2^{32}$  addition. A legal block occupies 80 busy cycles.

```
module sha1(clk_i, rst_i, text_i, text_o, cmd_i, cmd_w_i, cmd_o);
  input  clk_i, rst_i, cmd_w_i;
  input  [31:0] text_i;
  output [31:0] text_o;
  input  [3:0]  cmd_i;
  output [3:0]  cmd_o;
endmodule
```



Only `sha.v` may change. Reset behavior, command protocol, digest read order, latency, throughput and every observable cycle remain fixed. SHA-1 is a useful public vehicle for the method because it combines sequential control, word-level arithmetic and nonlinear Boolean logic; it is not recommended here as a security primitive.

### Corrected reference and provenance

The upstream artifact does not produce the standard digest for `abc`. A separate conformance correction created the golden source:

$$\text{SHA1}(\text{abc}) = \text{a9993e364706816aba3e25717850c26c9cd0d89d}.$$

That correction is historical preparation and is excluded from the claimed optimization. Every ratio in this report compares the accepted RTL with the corrected, frozen baseline.

| Artifact                              | SHA-256                                                          |
|---------------------------------------|------------------------------------------------------------------|
| Corrected baseline <code>sha.v</code> | 191a4f2148a4efda7aadd24480eb13d78a1d2c0c7e8a3fcc37c44f6a8e8011e5 |
| Accepted <code>sha.v</code>           | 743e6c9ffcca6f00d35d5e73ba6f6478a9133a0c55a471c16d6e59d831aeeabc |

### 3. Evaluation contract

#### FROZEN BEFORE MEASUREMENT

A PPA comparison is meaningful only when baseline and accepted RTL share the same functional, physical and statistical contract.

| Contract dimension         | Frozen choice                                                               |
|----------------------------|-----------------------------------------------------------------------------|
| <b>Editable artifact</b>   | <code>sha.v</code> only; every submission frozen by SHA-256.                |
| <b>Functional behavior</b> | Exact interface, reset, protocol, latency, output order and cycle behavior. |
| <b>Conformance</b>         | NIST SHA-1 Short and Long Message corpus; 129 cases.                        |
| <b>Formal</b>              | EQY commit <code>6734d8c2...</code> ; fail closed; 2 CPU, 7 GB, 10 min.     |
| <b>Physical flow</b>       | VTR/VPR commit <code>95f5c6de...</code> ; pinned Linux/amd64 image.         |
| <b>Architecture</b>        | <code>k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml</code> .                         |
| <b>Power</b>               | VTR PTM45, 0.9 V, 85 °C; fixed active and idle traces.                      |
| <b>PPA sample</b>          | 64 fixed, paired VPR seeds for certification.                               |
| <b>Primary metrics</b>     | Total MWTA, critical-path delay and energy per completed block.             |



Correctness precedes PPA. A failure, timeout or inconclusive formal result invalidates the source before any physical score can qualify it.

The RTL cannot alter testbenches, activity, architecture, tool flags, seeds or parsers. Acceptance belongs to the evaluator, not to the mechanism that proposed the source change.

## 4. Verification before PPA

### EVIDENCE FOR THE ACCEPTED RTL AND QUALIFICATION OF THE VERIFIER

| Gate               | What it protects                                                         | Status |
|--------------------|--------------------------------------------------------------------------|--------|
| Structural policy  | Interface, editable-file boundary, synthesizability and warning policy   | PASS   |
| Cycle regression   | Reset, protocol, state progression, latency and output order             | PASS   |
| NIST SHAVS         | 129 Short and Long Message SHA-1 conformance cases                       | PASS   |
| Sequential EQY     | All compared outputs for all cycles after reset under the declared model | PASS   |
| Physical integrity | Route, timing, power, hashes and exact seed pairing                      | PASS   |

### Formal scope

EQY compares public outputs and stable internal cut points. The proof is conservative and structure-aware: it can reject a legal but structurally distant redesign that does not converge within the laptop resource envelope. It supports local cycle-equivalent revisions. This conservatism does not weaken a reported pass: the accepted four-line revision was proved equivalent over the complete sequential circuit under the declared reset model.

### Mutation qualification

The 500 deterministic MCY mutations qualify the sensitivity of the verification stack. They are not a comparison across proposed candidates and not part of the baseline-versus-accepted PPA result.

| MCY outcome             | Count |
|-------------------------|-------|
| Detected by simulation  | 454   |
| Rejected only by formal | 28    |
| Proved equivalent       | 18    |
| Inconclusive            | 0     |

All 482 functionally distinguishable mutations were rejected: 454 by simulation and 28 only by formal. The remaining 18 were formally proved equivalent. This demonstrates that formal verification contributes detections beyond the functional corpus; mutation testing does not by itself prove absence of every possible defect.

## 5. Exact baseline-to-accepted diff

### FOUR LINES IN SHA.V

The line numbers are identical in the corrected baseline and accepted RTL. No register, state transition, port, command, latency or output statement changes.

```
sha.v:129 - assign SHA1_f1_BCD = (B & C) ^ (~B & D);
sha.v:129 + assign SHA1_f1_BCD = (B | D) & ((~B) | C);

sha.v:131 - assign SHA1_f3_BCD = (B & C) ^ (C & D) ^ (B & D);
sha.v:131 + assign SHA1_f3_BCD = (B & D) | (C & (B ^ D));

sha.v:141 - assign SHA1_Wt_1 = W13 ^ W8 ^ W2 ^ W0;
sha.v:141 + assign SHA1_Wt_1 = (W13 ^ W8) ^ (W2 ^ W0);

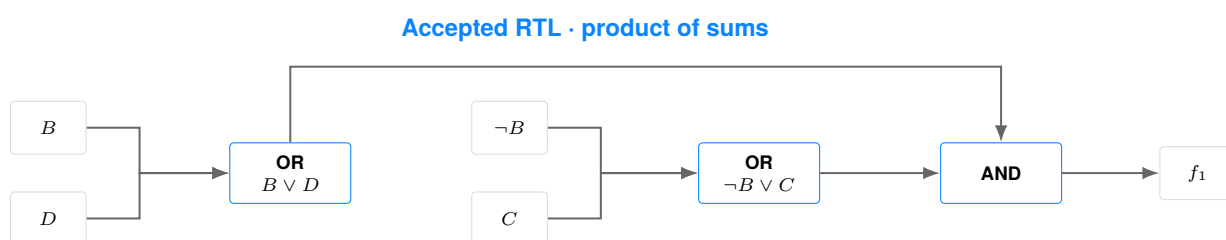
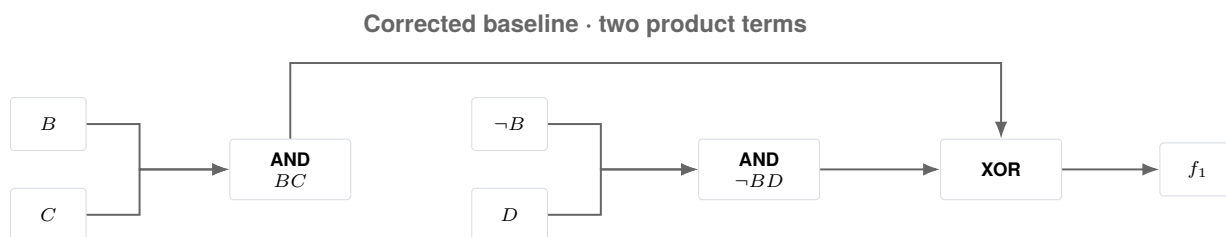
sha.v:144 - assign next_A = {A[26:0], A[31:27]}
           + SHA1_ft_BCD + E + Kt + Wt;
sha.v:144 + assign next_A = ({A[26:0], A[31:27]}
           + SHA1_ft_BCD) + (E + Kt + Wt);
```

**Attribution boundary.** The certified PPA result belongs to this combined four-line revision. The algebra below explains each equivalence and the conceptual topology exposed to synthesis. It does not assign an exact fraction of the measured PPA gain to any single line.



## 6. Rewrite 1: SHA-1 choose function

SOURCE: SHA.V:129



### Why the two expressions are equivalent

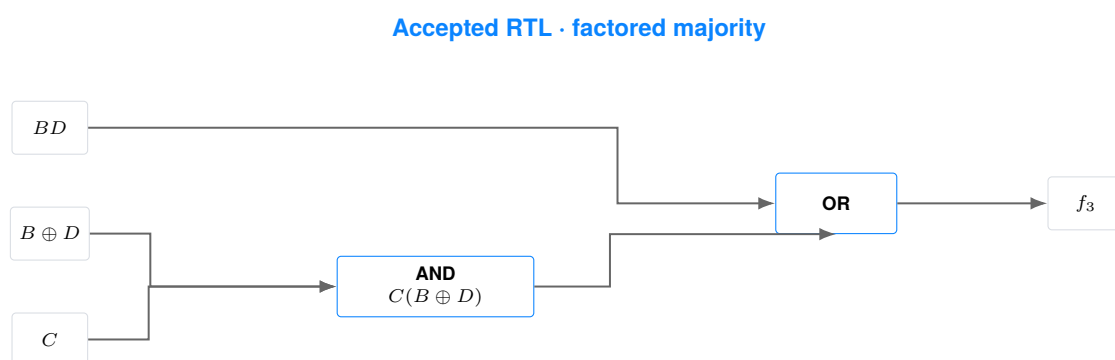
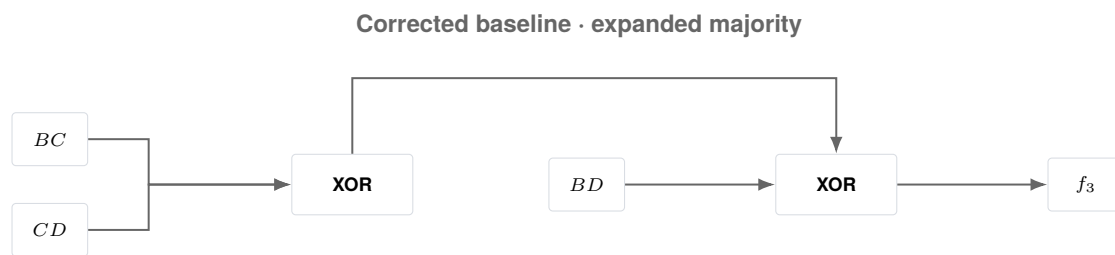
$$\begin{aligned}
 F &= BC \oplus (\neg B)D \\
 &= BC \vee (\neg B)D \\
 &= (B \vee D)((\neg B) \vee C).
 \end{aligned}$$

The products  $BC$  and  $(\neg B)D$  are mutually exclusive because  $B$  and  $\neg B$  cannot both be one, so their XOR equals OR. Expanding the accepted product-of-sums gives  $BC \vee (\neg B)D \vee CD$ . The  $CD$  consensus term is redundant: when  $C = D = 1$ , either  $B = 1$  makes  $BC$  true or  $B = 0$  makes  $(\neg B)D$  true. Both forms therefore select  $C$  when  $B$  is one and  $D$  otherwise.

**Formal result.** EQY proved this identity in the complete sequential circuit. The diagram is conceptual; the FPGA mapper remains free to choose the final LUT implementation.

## 7. Rewrite 2: SHA-1 majority function

SOURCE: SHA.V:131



### Why the two expressions are equivalent

$$\begin{aligned}
 M &= BC \oplus CD \oplus BD \\
 &= BD \oplus C(B \oplus D) \\
 &= BD \vee C(B \oplus D).
 \end{aligned}$$

The first two lines are the same XOR polynomial after factoring  $C$  from  $BC \oplus CD$ . The terms  $BD$  and  $B \oplus D$  cannot both be one: if  $BD = 1$ , then  $B = D = 1$  and  $B \oplus D = 0$ . Their XOR can therefore be replaced by OR. If  $B$  equals  $D$ , their shared value decides the majority; if they differ,  $C$  decides it.

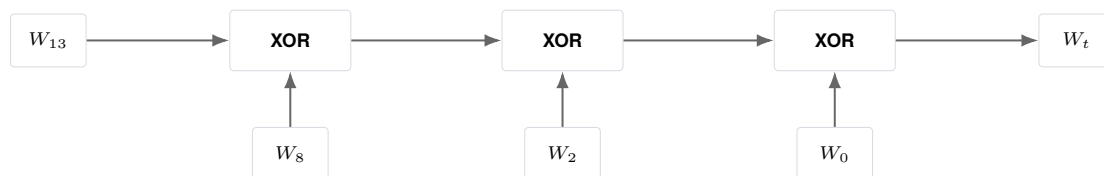
**Formal result.** EQY proved the actual Verilog implementations equivalent over the complete sequential design.



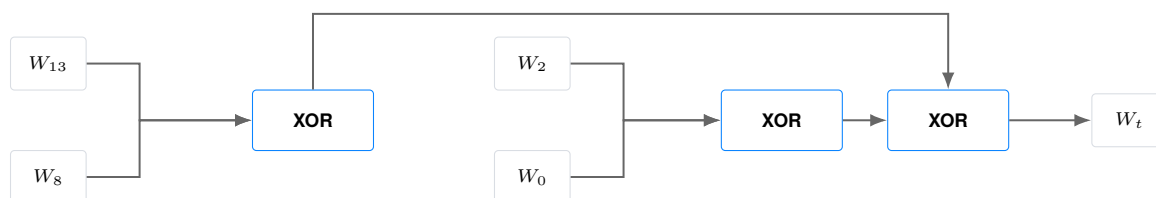
## 8. Rewrite 3: balanced message-schedule XOR

SOURCE: SHA.V:141

Corrected baseline · serial XOR chain



Accepted RTL · balanced XOR tree



### Why the two expressions are equivalent

$$\begin{aligned}
 X &= W_{13} \oplus W_8 \oplus W_2 \oplus W_0 \\
 &= (W_{13} \oplus W_8) \oplus (W_2 \oplus W_0).
 \end{aligned}$$

Bitwise XOR is associative at every bit position, so regrouping changes no result bit and introduces no carry. The parentheses expose two parallel first-level operations before the final XOR. A synthesizer may rebalance the baseline itself, but explicit grouping can influence intermediate optimization, LUT packing and routing topology.

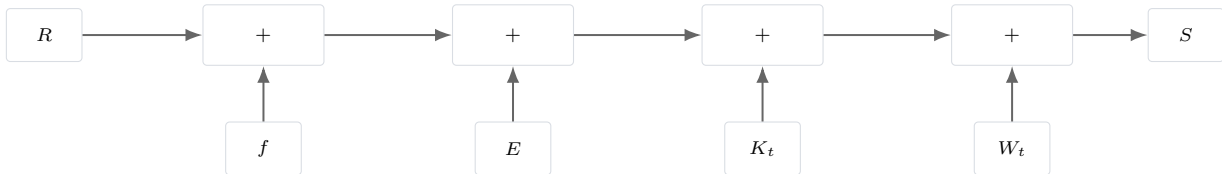
**Formal result.** EQY proved the same observable state sequence for the actual sequential circuit.

## 9. Rewrite 4: reassocated round accumulator

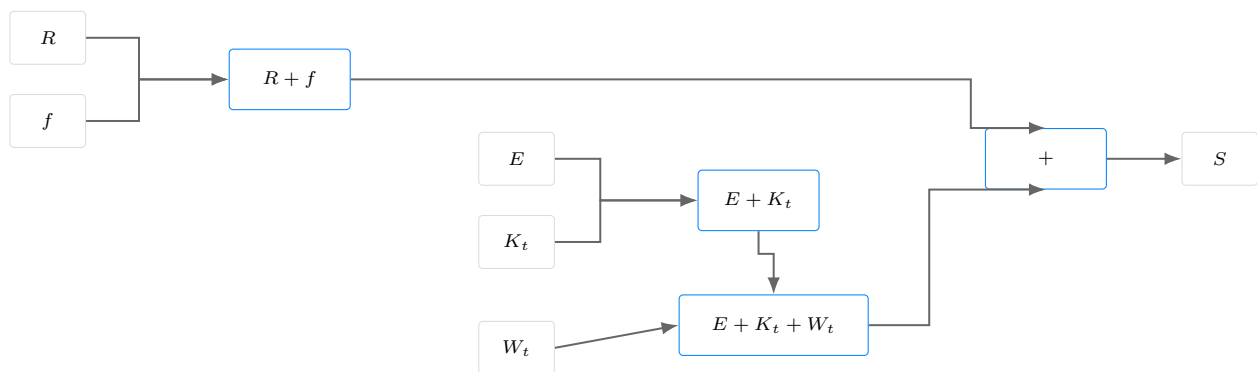
SOURCE: SHA.V:144

Here  $R$  denotes rotate-left-by-5 of  $A$ , and  $f$  denotes `SHA1_ft_BCD`.

Corrected baseline · serial accumulation



Accepted RTL · grouped dependency tree



### Why the two expressions are equivalent

$$\begin{aligned}
 S &= (R + f + E + K_t + W_t) \bmod 2^{32} \\
 &= ((R + f) + (E + K_t + W_t)) \bmod 2^{32}.
 \end{aligned}$$

Every operand and the assigned result is a 32-bit unsigned vector. Addition is associative modulo  $2^{32}$ : discarding an overflow carry at an intermediate grouping cannot change the final low 32 bits. The accepted form changes the exposed dependency structure, not the numerical result. This identity depends on the declared widths and unsigned vector semantics.

**Formal result.** EQY confirmed the actual Verilog interpretation over the complete sequential design.

## 10. PPA environment and statistics

### OPEN 45 NM FPGA PROXY

The target is VTR's homogeneous `k6_N10_I40_Fi6_L4_frac0_ff1_45nm` architecture: six-input LUTs clustered ten per logic block, with the architecture's routing and timing model. Power uses VTR PTM45 properties at 0.9 V and 85 °C. MWTA is a minimum-width-transistor-area abstraction, not physical square-micrometer area.

Two 5,000-cycle ACE traces are frozen and hashed. The active trace continuously executes diverse legal blocks and completes 60 blocks; the idle trace keeps the clock active after reset with an inactive interface. Both power analyses reuse the same placement and route.

$$E_{\text{block}} = \frac{P_{\text{active,total}} t_{\text{routed workload}}}{60 \text{ completed blocks}}.$$

Baseline and accepted RTL use the same 64 VPR seeds. Ratios are analyzed in log space because PPA comparisons are multiplicative. For seed  $s$ :

$$r_s = \sqrt[3]{r_{A,s} r_{D,s} r_{E,s}}, \quad \hat{r} = \exp\left(\frac{1}{64} \sum_{s=1}^{64} \log r_s\right).$$

Two-sided 95% Student- $t$  intervals use 63 degrees of freedom. The baseline is exactly 1.0 in paired-ratio space by construction, so baseline and accepted seed clouds are instead normalized to the same baseline median in the next figure. This preserves baseline variability and makes the physical displacement of the accepted cloud visible on one axis.

**Decision rule.** Every correctness and integrity gate must pass; at least one primary metric must improve materially; no primary metric may show statistical evidence of regression; and the composite one-sided 95% upper bound must remain below 1.0.

# 11. Every paired certification seed

## DISTRIBUTION, NOT A SINGLE IMPLEMENTATION

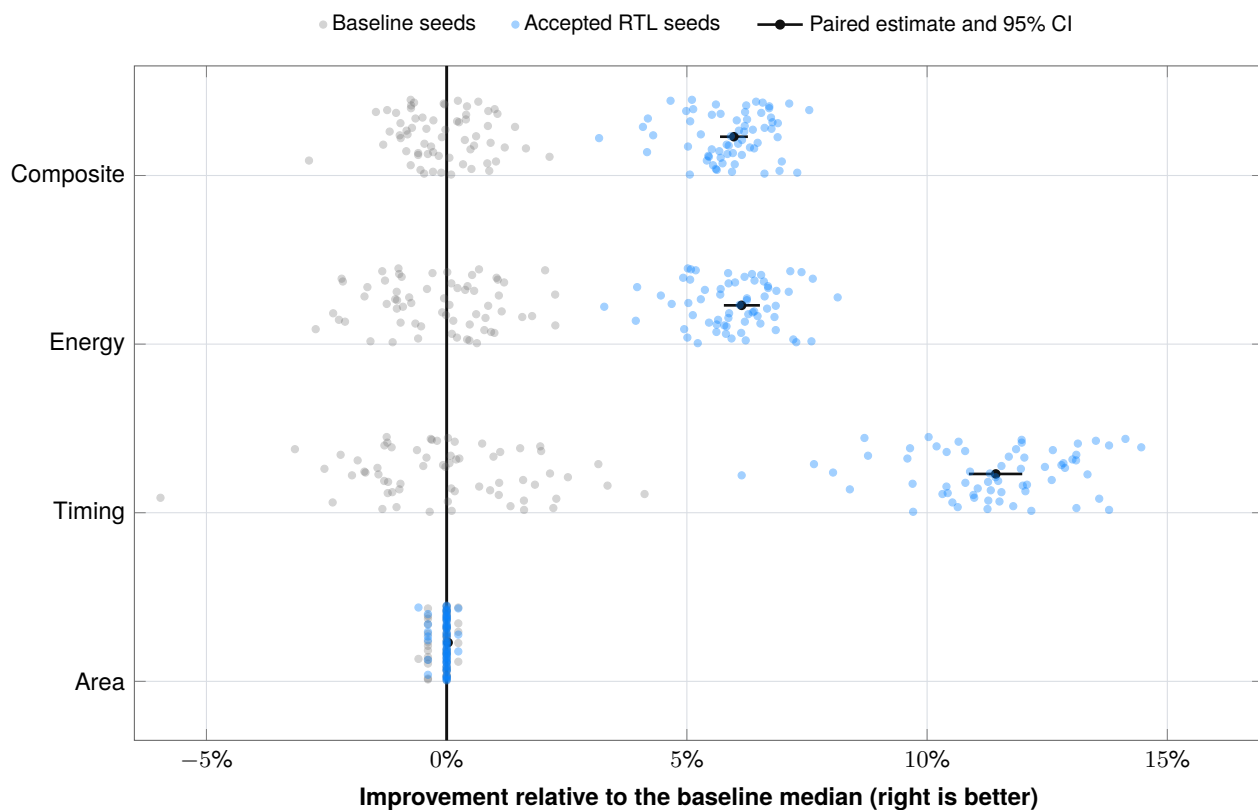


FIGURE 2 · OVERLAID BASELINE AND ACCEPTED SEED CLOUDS

Gray and blue points are absolute seed results normalized to the same baseline median; the same seed receives the same vertical jitter, so horizontal displacement reveals the change in implementation. Black points and segments retain the formal same-seed paired estimate and 95% interval. The clouds explain physical variability; the black paired statistic determines confidence.

| Metric    | Wins | Ties | Losses |
|-----------|------|------|--------|
| Area      | 14   | 41   | 9      |
| Timing    | 64   | 0    | 0      |
| Energy    | 64   | 0    | 0      |
| Composite | 64   | 0    | 0      |

## 12. Power, energy and physical evidence

### WHY LOWER ENERGY CAN COEXIST WITH HIGHER POWER

| Informative metric   | Baseline median | Accepted median | Raw median change |
|----------------------|-----------------|-----------------|-------------------|
| Fmax                 | 66.6426 MHz     | 75.2964 MHz     | +12.99%           |
| Active total power   | 9.9125 mW       | 10.4900 mW      | +5.83%            |
| Active dynamic power | 5.0713 mW       | 5.7018 mW       | +12.43%           |
| Active static power  | 4.8394 mW       | 4.7916 mW       | −0.99%            |
| Idle total power     | 4.4670 mW       | 4.6550 mW       | +4.21%            |
| Energy / block       | 12.3896 nJ      | 11.6399 nJ      | −6.05%            |

The accepted RTL consumes more active power per unit time but less modeled energy per completed block because it finishes the workload sooner. A product constrained by peak power, thermal density or idle power could use a different acceptance policy.

### Representative post-route implementation: seed 20

| Post-route evidence | Baseline | Accepted RTL | Observed change |
|---------------------|----------|--------------|-----------------|
| ABC .names nodes    | 1643     | 1652         | +9              |
| Packed CLBs         | 188      | 187          | −1              |
| Timing-graph levels | 46       | 42           | −4              |

For seed 20, the accepted RTL has nine more ABC .names nodes (1643 to 1652), packs into one fewer CLB (188 to 187) and reduces the timing graph from 46 to 42 levels. This rules out the simplistic explanation that fewer generic mapped nodes caused the result. The supported interpretation is that the changed topology gives VTR a more favorable packing and routing solution. Seed 20 illustrates mechanism; the 64 paired implementations establish repeatability.

## 13. Reproducibility and evidence identity

### FROM CLEAN CHECKOUT TO THE SAME DECISION

The fast offline audit needs Python 3 only:

```
make verify
```

It checks RTL identities, formal and NIST references, exact 64-seed pairing, metric log-ratio estimates, confidence intervals and the acceptance conditions.

The full pinned rerun starts with:

```
./reproduce/build-image.sh
./reproduce/run-candidate.sh \
  rtl/accepted/sha.v \
  certification \
  /absolute/new/results/certification \
  accepted-rtl-reproduction
```

The Linux/amd64 container is capped at two CPUs, 7 GB RAM and 512 processes with runtime networking disabled. The certified run completed in 1,091.82 seconds on the qualified Apple Silicon laptop. Host contention affects wall time, not source hashes, seeds or parsed implementation results.

| Artifact                 | SHA-256                                                          |
|--------------------------|------------------------------------------------------------------|
| Corrected baseline sha.v | 191a4f2148a4efda7aadd24480eb13d78a1d2c0c7e8a3fcc37c44f6a8e8011e5 |
| Accepted sha.v           | 743e6c9ffcca6f00d35d5e73ba6f6478a9133a0c55a471c16d6e59d831aeeabc |
| EQY PASS marker          | ba3b47c5fbb189844a827ae395e816024c967b83444a06ecb71f9e34498ab07  |
| EQY driver log           | 203f2ab1a1db8aadcbdbf5be88e5478b1abcb51e213e941557889e1474b9cbce |
| Certification archive    | 9983b1fef4509b9a9a592af8134be39eaa7545e5269ac7332206e86db7cce3e8 |

The complete certification archive is distributed separately from normal Git history. Commands, logs, resource usage, tool versions, seeds, hashes and per-run results are retained as evidence.

# 14. Limits and customer transfer

## WHAT THIS RESULT DOES AND DOES NOT ESTABLISH

This report does not provide:

- ASIC synthesis, place-and-route or signoff;
- commercial-FPGA implementation data;
- extracted parasitics, PVT corners, clock-tree signoff, IR drop, EM, DRC/LVS, package, yield or production-test analysis;
- proof that the same source edits improve another architecture, technology or workload;
- exact PPA attribution to an individual rewrite;
- a recommendation to use SHA-1 in a new security design.

The 64-seed confidence interval models implementation-seed variability. It does not model process, voltage, temperature, workload or tool-version uncertainty.

**Transfer to a customer pilot.** Replace the academic proxy with customer-owned RTL, formal strategy, libraries, constraints, activity, tools and signoff policy. The transferable capability is the combination of a small inspectable source change and an evaluator that can justify acceptance to the engineers responsible for the system.

## Primary references

- [VTR/VPR documentation](#)
- [VTR power estimation](#)
- [EQY sequential equivalence checking](#)
- [NIST secure hashing validation](#)

## Case study and contact

- [Reproducible case-study repository](#)
- [Göther Labs · evaluation-driven optimization](#)

**Conclusion.** Within the declared VTR 45 nm FPGA proxy, the accepted four-line RTL revision is cycle-equivalent to the corrected baseline and improves the certified composite PPA estimate by 5.98%. The result is inspectable, reproducible and explicitly bounded to the evaluation environment.